SQL SERVER DATABASE ADMINISTRATION

45-Day Online Training Syllabus

Training Mode	Live online sessions
Duration	45 days
Daily Session	1 hour 30 minutes
Total Instructor-Led Hours	67.5 hours
Level	Beginner to Intermediate Foundations
Methodology	Concepts, live demonstrations, hands-on labs, troubleshooting, and assessments

Training Objective

Build a practical foundation in SQL Server administration, including installation, database management, security, backup and recovery, automation, monitoring, performance troubleshooting, and high availability and disaster recovery fundamentals.

Daily Session Structure

Activity	Duration
Concepts and explanation	20 minutes
Live demonstration	25 minutes
Hands-on practice	35 minutes
Troubleshooting and Q&A	10 minutes
Total	**90 minutes**

Target audience: Freshers, SQL developers transitioning to DBA roles, IT professionals entering database administration, and aspiring SQL Server DBAs.

Syllabus Overview

Module	Coverage	Days
1. Fundamentals and Installation	Architecture, installation, SSMS, services, system databases	1–7
2. Database Administration and Storage	Database creation, files, recovery models, metadata	8–14
3. Security, Backup, and Recovery	Logins, permissions, backup types, restore, point-in-time recovery	15–21
4. SQL Server Agent and Maintenance	Jobs, notifications, integrity, indexes, statistics, TempDB	22–28
5. Monitoring and Troubleshooting	DMVs, CPU/memory/I/O, blocking, deadlocks, query plans, incidents	29–35
6. HA/DR and DBA Operations	Log shipping, AGs, FCIs, patching, migration, operational checks	36–42
7. Scenarios and Assessment	Incident simulation, practical assessment, interview preparation	43–45

Module 1: SQL Server Fundamentals and Installation (Days 1–7)

Day 1: Introduction to SQL Server DBA

- Introduction to SQL Server and RDBMS concepts.
- Roles and responsibilities of a SQL Server DBA.
- Development, test, staging, and production environments.
- Daily activities and responsibilities of a DBA.
- Overview of database administration career opportunities.

Day 2: SQL Server Architecture

- SQL Server instances and database engine.
- Relational Engine and Storage Engine fundamentals.
- SQL Server services and client connectivity.
- SQL Server memory and transaction processing basics.
- SQL Server editions and versions.

Day 3: SQL Server Installation

- Installation prerequisites and hardware requirements.
- SQL Server edition selection and installation options.
- Default and named instances.
- Authentication modes and service accounts.
- SQL Server installation and configuration basics.

Day 4: SQL Server Management Studio (SSMS)

- SSMS installation and connectivity.
- Object Explorer and Query Editor.
- Activity Monitor and SQL Server error logs.
- SQL Server Configuration Manager.
- Basic T-SQL administration commands.

Day 5: System Databases and Database Files

- master, model, msdb, and tempdb.
- MDF, NDF, and LDF files.
- Database pages and extents.
- Data storage and transaction log fundamentals.
- Physical and logical database file names.

Day 6: SQL Server Services and Connectivity

- SQL Server Database Engine service.
- SQL Server Agent service.
- Windows services and service accounts.
- TCP/IP, ports, and named-instance connectivity.
- Basic connection failure troubleshooting.

Day 7: Architecture Revision and Practical Assessment

- Revision of installation and architecture concepts.
- Instance and database identification.
- System database exploration.
- SQL Server environment inventory.
- Basic DBA troubleshooting exercises.

Module 2: Database Administration and Storage (Days 8–14)

Day 8: Database Creation and Management

- Creating databases using SSMS and T-SQL.
- CREATE DATABASE, ALTER DATABASE, and DROP DATABASE.
- Database properties and configuration.
- Database naming conventions.
- Database creation troubleshooting.

Day 9: Data Files, Log Files, and Filegroups

- Primary and secondary data files.
- Transaction log files.
- Creating and managing filegroups.
- File placement and storage planning.
- Database file management.

Day 10: Database Sizing and Autogrowth

- Initial database file size.
- File growth increments and maximum size.
- Percentage-based vs. fixed-size autogrowth.
- Database capacity planning.
- Disk space and database growth monitoring.

Day 11: Database Recovery Models

- SIMPLE recovery model.
- FULL recovery model.
- BULK_LOGGED recovery model.
- Transaction log behavior and truncation.
- Recovery model selection and operational impact.

Day 12: Database States and Options

- ONLINE, OFFLINE, RESTORING, SUSPECT, and RECOVERY_PENDING states.
- SINGLE_USER and MULTI_USER modes.
- READ_ONLY and READ_WRITE databases.
- Database compatibility level.
- Database state troubleshooting.

Day 13: Database Metadata and System Views

- sys.databases.
- sys.database_files.
- sys.master_files.
- Database inventory and configuration queries.
- Retrieving database status, size, and recovery model.

Day 14: Database Administration Practical

- Creating and configuring a database.
- Managing data and log files.
- Configuring recovery models.
- Reviewing database properties.
- Practical database administration assessment.

Module 3: Security, Backup, and Recovery (Days 15–21)

Day 15: SQL Server Security Fundamentals

- Authentication vs. authorization.
- Windows Authentication and SQL Server Authentication.
- Server logins and database users.
- Security principals.
- Principle of least privilege.

Day 16: Roles and Permissions

- Fixed server roles.
- Database roles.
- GRANT, DENY, and REVOKE.
- Schema-level and object-level permissions.
- User access validation and troubleshooting.

Day 17: Backup Fundamentals and Strategy

- Importance of database backups.
- Full, differential, transaction log, and copy-only backups.
- Recovery Point Objective (RPO).
- Recovery Time Objective (RTO).
- Backup retention and scheduling strategy.

Day 18: Full and Differential Backups

- Backup destinations and permissions.
- Performing full database backups.
- Performing differential backups.
- Backup compression and checksums.
- Backup history and verification.

Day 19: Transaction Log Backups

- Transaction log architecture.
- Transaction log backup chain.
- Log truncation and log reuse waits.
- Active transactions and log growth.
- Transaction log backup troubleshooting.

Day 20: Database Restore and Recovery

- Restore sequence and recovery process.
- Restoring full and differential backups.
- Restoring transaction log backups.
- WITH NORECOVERY and WITH RECOVERY.
- Database relocation using WITH MOVE.

Day 21: Point-in-Time Recovery and Validation

- Point-in-time recovery using STOPAT.
- RESTORE VERIFYONLY.
- Backup checksums and backup metadata.
- Test restores and recovery validation.
- Backup and restore practical assessment.

Module 4: SQL Server Agent and Database Maintenance (Days 22–28)

Day 22: SQL Server Agent Fundamentals

- SQL Server Agent architecture.
- Jobs, job steps, and schedules.
- Job owners and execution context.
- Job history and activity monitoring.
- Creating basic automated DBA jobs.

Day 23: Job Scheduling and Troubleshooting

- One-time and recurring schedules.
- Job failures and retry settings.
- Permissions and job ownership.
- Job history and error analysis.
- Troubleshooting jobs that do not execute.

Day 24: Database Mail and Notifications

- Database Mail architecture.
- SMTP configuration fundamentals.
- Mail profiles and accounts.
- SQL Server Agent operators.
- Job failure email notifications.

Day 25: Database Integrity Checks

- Database consistency and integrity.
- DBCC CHECKDB.
- DBCC CHECKTABLE.
- Understanding consistency-check errors.
- Scheduling database integrity checks.

Day 26: Index Fundamentals and Maintenance

- Clustered and nonclustered indexes.
- Index fragmentation.
- Index rebuild vs. reorganize.
- Index maintenance considerations.
- Monitoring index health.

Day 27: Statistics and Maintenance Automation

- SQL Server statistics fundamentals.
- Automatic statistics updates.
- UPDATE STATISTICS.
- Maintenance windows and job scheduling.
- Automating routine maintenance tasks.

Day 28: TempDB Administration

- TempDB architecture and usage.
- TempDB data and log files.
- File sizing and autogrowth.
- TempDB contention fundamentals.
- TempDB space usage monitoring.

Module 5: Monitoring, Performance, and Troubleshooting (Days 29–35)

Day 29: SQL Server Monitoring and DMVs

- Dynamic Management Views (DMVs).
- Sessions, requests, and connections.
- Database status monitoring.
- Current workload inspection.
- Building a basic DBA health-check report.

Day 30: CPU, Memory, and Disk I/O

- CPU utilization and resource pressure.
- SQL Server memory configuration.
- Buffer pool and Page Life Expectancy.
- Disk I/O latency and storage performance.
- Windows performance monitoring fundamentals.

Day 31: Blocking and Transaction Management

- Locks and blocking fundamentals.
- Identifying blocked sessions and lead blockers.
- Blocking chains and long-running transactions.
- Isolation levels and lock escalation.
- Blocking troubleshooting techniques.

Day 32: Deadlocks and Extended Events

- Deadlock concepts and causes.
- Deadlock victim selection.
- Reading deadlock graphs.
- Introduction to Extended Events.
- Deadlock prevention and troubleshooting.

Day 33: Slow Queries and Execution Plans

- Query execution plan fundamentals.
- Actual vs. estimated execution plans.
- Index scans, index seeks, and key lookups.
- Logical reads and query performance.
- Basic query optimization techniques.

Day 34: Transaction Log and Disk-Space Incidents

- Transaction log full errors.
- Log reuse waits and active transactions.
- Missing transaction log backups.
- Disk space shortage and autogrowth issues.
- Safe troubleshooting and corrective actions.

Day 35: Error Logs and Production Troubleshooting

- SQL Server error logs.
- Login failure investigation.
- Service startup issues.
- Failed backups and SQL Agent jobs.
- Incident documentation and root cause analysis.

Module 6: High Availability, Disaster Recovery, and DBA Operations (Days 36–42)

Day 36: High Availability and Disaster Recovery

- HA vs. DR.
- RPO and RTO planning.
- Failover and recovery concepts.
- Business continuity planning.
- Disaster recovery testing fundamentals.

Day 37: Log Shipping

- Log shipping architecture.
- Primary and secondary databases.
- Backup, copy, and restore jobs.
- Monitoring and alerting.
- Failover procedures and limitations.

Day 38: Always On Availability Groups

- Availability Group architecture.
- Primary and secondary replicas.
- Availability databases.
- Synchronous vs. asynchronous commit.
- Automatic and manual failover.

Day 39: Failover Cluster Instances and HA/DR Comparison

- Windows Server Failover Clustering fundamentals.
- Shared storage and instance-level availability.
- Failover Cluster Instance architecture.
- FCI vs. Availability Groups vs. log shipping.
- HA/DR solution selection.

Day 40: SQL Server Patching and Upgrades

- SQL Server version and build identification.
- Cumulative updates and security updates.
- Compatibility levels and upgrade planning.
- Pre-upgrade checks and rollback planning.
- Post-patching validation.

Day 41: Database Migration Fundamentals

- Backup/restore migration approach.
- Detach/attach considerations.
- Database compatibility and application validation.
- Migration of logins, SQL Agent jobs, and dependencies.
- Migration checklist and post-migration verification.

Day 42: Daily, Weekly, and Monthly DBA Operations

- Daily database health checks.
- Backup and job failure reviews.
- Disk space and database growth monitoring.
- Security and maintenance reviews.
- Capacity planning and operational documentation.

Module 7: Real-Time DBA Scenarios and Final Assessment (Days 43–45)

Day 43: Production Incident Simulation

- Investigating slow database performance.
- Identifying blocking and long-running queries.
- Diagnosing transaction log growth.
- Troubleshooting failed jobs and backups.
- Root cause analysis and incident documentation.

Day 44: End-to-End DBA Practical Assessment

- Database creation and configuration.
- Security and permission management.
- Backup and restore operations.
- SQL Server Agent automation.
- Database integrity and health checks.
- Monitoring and troubleshooting assessment.

Day 45: Final Revision and Interview Preparation

- SQL Server DBA fundamentals revision.
- Frequently asked DBA interview questions.
- Real-time troubleshooting discussions.
- Review of practical assignments.
- Roadmap to intermediate and advanced DBA skills.

Training Methodology

- Daily 90-minute instructor-led online session.
- Concept explanation, live demonstrations, guided practical labs, and Q&A.;
- Weekly revision and practical assessments.
- Final production-style troubleshooting exercise.

Expected Learning Outcomes

- Install and configure SQL Server and use SSMS for administration.
- Create and configure databases, files, and transaction logs.
- Manage logins, users, roles, and permissions.
- Perform backup, restore, and point-in-time recovery exercises.
- Automate routine tasks with SQL Server Agent.
- Perform database integrity checks and basic maintenance.
- Monitor SQL Server and investigate common performance issues.
- Identify blocking, deadlocks, and transaction log problems.
- Explain HA/DR architectures and database migration fundamentals.
- Follow a structured production incident investigation process.

Lab Requirements

- Windows computer or virtual machine.

- SQL Server Developer Edition for non-production learning.
- SQL Server Management Studio (SSMS).
- Recommended 8 GB RAM minimum; 16 GB is preferable.
- Approximately 50–100 GB free disk space, depending on installation and lab needs.
- Additional Windows Server/infrastructure may be needed for advanced HA/DR labs; architecture demonstrations can be used if unavailable.

Note: This is a foundational training program and does not guarantee independent production readiness. Students should continue with supervised practical experience and advanced training in performance tuning, HA/DR implementation, security, patching, and migrations.